

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2021

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
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- Let G be a group of order $7 \times 43 \times 47$ and suppose the center of G contains an element of order 7. Show that G is abelian.
- Let $f(x) = (x-1)(x-2)^2(x-3)^3$ and let $V = \mathbb{C}[x]/(f(x))$, regarded as a vector space over $\mathbb{C}$. Let T be the linear transformation defined by multiplication by x ; that is, $T(g(x)) = xg(x) \pmod{f(x)}$. Find the Jordan canonical form for T .
- Let R be a commutative Noetherian ring with 1 and let M be an R -module. Let $0 \neq x \in M$ and let $I = \{r \in R \mid rx = 0\}$. Show that there exists a prime ideal P of R with $P \supseteq I$ and an element $y \neq 0$ in M such that $P = \{r \in R \mid ry = 0\}$. (*Hint:* Consider the set of ideals $J \supseteq I$ such that J is the annihilator of some nonzero element of M .)
- (a) Let R be a commutative ring with 1 and let M be a finitely generated R -module. Suppose there exists a finitely generated projective R -module P and a surjective R -homomorphism $\phi : M \rightarrow P$. Show that the kernel of ϕ is a finitely generated R -module.
(b) Give an explicit example of a ring R (commutative with 1) and a surjection $\psi : M \rightarrow N$ of finitely generated R -modules such that the kernel of ψ is not a finitely generated R -module. (*Hint:* If I is an ideal, $R \rightarrow R/I$ is surjective.)
- Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree 4, and let $r_1, r_2, r_3, r_4 \in \mathbb{C}$ be the roots of $f(x)$. Let

$$s_1 = r_1r_2 + r_3r_4, \quad s_2 = r_1r_3 + r_2r_4, \quad s_3 = r_1r_4 + r_2r_3.$$

Let $g(x) = (x - s_1)(x - s_2)(x - s_3)$.

- Show that $g(x) \in \mathbb{Q}[x]$.
- Let K be the splitting field of $f(x)$ and let $G = \text{Gal}(K/\mathbb{Q})$. Show that if G has order 4 then $g(x)$ has a root in $\mathbb{Q}$.

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6. Suppose the group G has the following character table (A, B, C, D, E, F are the conjugacy classes in G , and $\chi_0, \dots, \chi_5$ are the irreducible characters of G):

	A	B	C	D	E	F
χ_0	1	1	1	1	1	1
χ_1	1	-1	1	1	-1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	1	1	-1	-1	-1
χ_4	2	0	-1	2	0	-1
χ_5	2	0	-1	-2	0	*

- (a) What is the missing number? (denoted by *)
- (b) Show that G has no normal subgroup H with $[G : H] = 3$.
- (c) Show that G has a normal subgroup K with $[G : K] = 4$.

(Hint for parts (b) and (c): One-dimensional characters are the homomorphisms from G to $\mathbb{C}^\times$.)